

Particulate Matter Monitoring & Health Effects

Issue 2026-08-12 | Entry window 12 August 2026 | 28 records carried, 14 rejected

28RECORDS CARRIED
(14 REJECTED)**8**SUBTOPIC CLUSTERS
REPRESENTED**5**EFFECT ESTIMATES
WITH INTERVALS**11**INSTRUMENTATION /
EXPOSURE RECORDS**6**METADATA-ONLY
(NO ABSTRACT)

Signal of the day

The Parkinson's-PM_{2.5} association is carried by exposure at least four years before the event — which makes the exposure-history requirement, not the concentration, the binding constraint on the design.

Delaney et al. (*Environ Epidemiol*) fitted distributed-lag discrete-time survival models to **10,366,083 Medicare fee-for-service beneficiaries** aged 65+ across the contiguous US, 2000–2016, with 10-year exposure histories for PM_{2.5}, NO₂ and summer ozone. Ten years of continuous PM_{2.5} at the 90th versus the 0.5th percentile (11.8 vs. 3.0 $\mu\text{g}/\text{m}^3$) gave **OR 1.634 (95% CI 1.489–1.792)** for first hospitalisation with PD; NO₂ at 31.7 vs. 3.7 ppb gave **1.474 (1.379–1.575)**. Ozone was equivocal. The load-bearing detail is the lag structure: associations were carried by exposure *at least four years* before hospitalisation, with the near-event years contributing little.

Two consequences follow for instrumentation. First, the exposure contrast that matters here is well below current US design values — 11.8 $\mu\text{g}/\text{m}^3$ is an unremarkable annual mean — so the measurement problem is long-run bias and drift, not peak capture. A network whose calibration wanders by 1–2 $\mu\text{g}/\text{m}^3$ over a multi-year deployment degrades exactly the contrast this model exploits. Second, hospitalisation with PD is a proxy for onset that is displaced by an unknown and probably differential interval; the four-year lag should be read as a lower bound on latency, not an estimate of it.

What changed today

- **Emission-factor choice is invisible in total mass and decisive in source attribution.** Park et al. ran EMEP/EEA and KIMM non-exhaust factors through the same 3-D CFD–chemistry model over a dense Seoul district. Total non-exhaust PM₁₀ was similar under both; the partitioning was not — brake wear dominant under EMEP/EEA, road wear dominant under KIMM. Non-exhaust reached **up to 70% of total PM₁₀** in near-road cells. A monitor reporting mass alone cannot distinguish the two worlds, and they imply opposite mitigation priorities.
- **Two independent mediation results converge on oxidative/inflammatory metabolism.** Chow et al. (*Nat Commun*) profiled >1,100 pre-diagnostic plasma metabolites in 1,357 Cancer Prevention Study participants and found eight metabolites associated with both exposure and subsequent lung cancer; four (γ -glutamylglutamine, phenylacetylglutamate, N-(2-furoyl)glycine, 4-vinylguaicol glucuronide) mediated the PM and ozone associations at $q < 0.2$. On the same day, a correspondence in *JACC Adv* pushes back on the statistical fragility of exactly this class of metabolomic mediation analysis — worth reading as a pair.
- **Renal burden is now quantified with a pooled exposure–response function.** Li et al. pooled 38 CKD and 16 eGFR studies: **RR 1.32 (1.21–1.43) per 10 $\mu\text{g}/\text{m}^3$** , predominantly log-linear, with a saturating eGFR response at high concentration. Attributed 2020 burden is 485,000 deaths and 243 million prevalent cases; PM_{2.5} ranks second among level-2 CKD mortality risk factors, up from first in 1990 only because metabolic risks grew faster.
- **Instrumentation: a non-flammable CPC working fluid reaches parity.** Kirchhoff et al. ran a DMSO-based condensation particle counter against a butanol counterpart for six months. Mean cutoff **5.8±0.9 nm (DMSO) vs. 5.6±0.5 nm (butanol)**, with substantially lower fluid consumption. The practical unlock is airborne and unattended remote deployment, where butanol's flammability is the blocking constraint.
- **Open-source exposure tooling.** Chillrud et al. released *pcpr*, an R implementation of principal component pursuit with three environmental-health extensions (non-negativity, missingness, below-LOD penalty). Applied to 26 PM_{2.5} constituents in Queens 2015–2021 it recovered four patterns and 211 outlying events including July Fourth firework spikes. Directly usable on speciated network output.
- **The abstract famine continues — fifth consecutive issue.** Six records carried today have no abstract at the entry date: four Elsevier *Atmos. Environ.* articles, one Elsevier correspondence, one Lancet editorial. The four *Atmos. Environ.* papers are squarely in scope (mobile monitoring, São Paulo PM₁₀ speciation, Australian biomass-burning fingerprints, absorbing aerosol index) and carry no extractable content. They are queued for the Saturday weekly sweep.

Corpus analytics

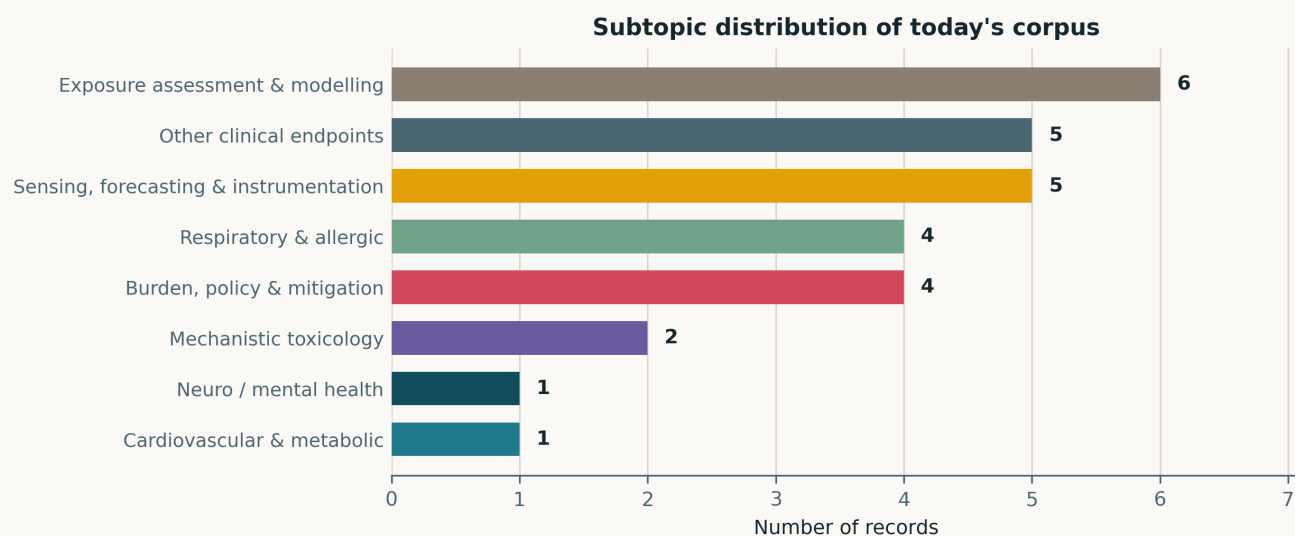


Fig. Subtopic assignment of the 28 carried records, single-label by dominant contribution. Exposure assessment and sensing together account for 11 records (39%), the highest instrumentation share since 3 August; the balance is unusually review-heavy on the health side. Neither *Reproductive & developmental* nor *Occupational & indoor* received a record today.

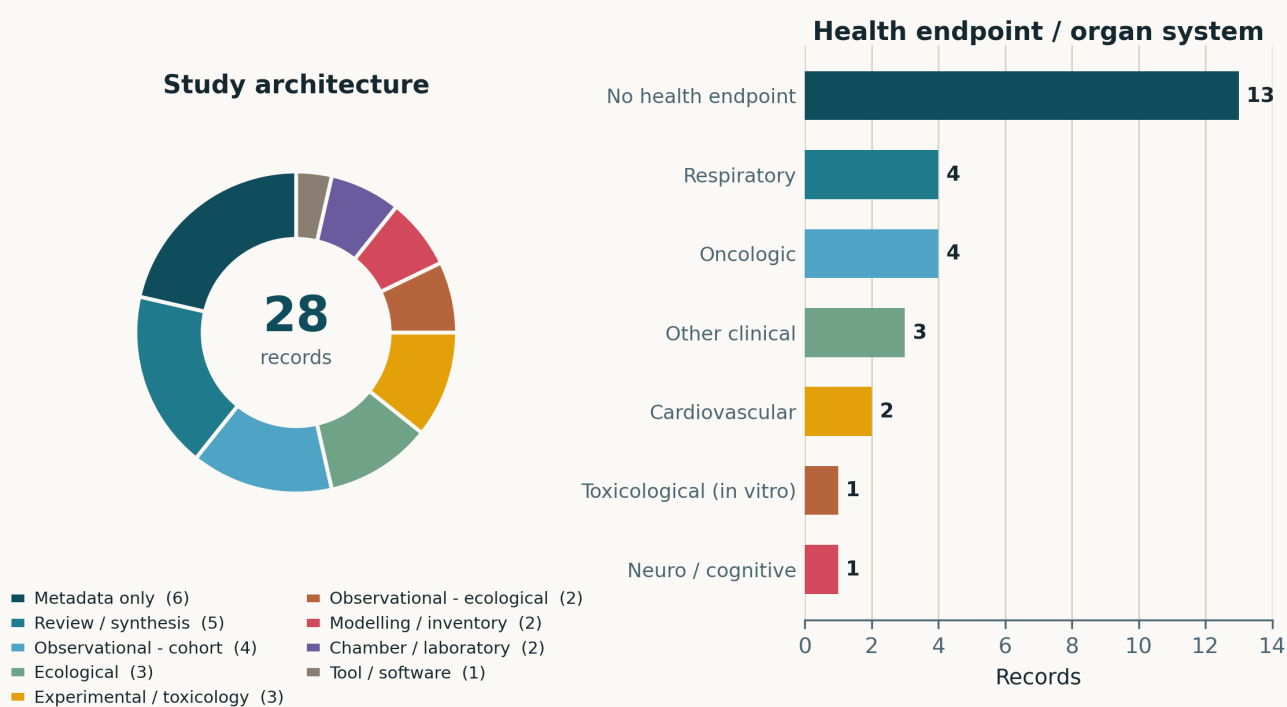


Fig. Left — study architecture. The *Metadata only* slice (6) is reported as its own category rather than pooled with characterised designs; it is a property of the harvest, not of the science. Right — health endpoint. Twelve records carry no human health endpoint at all, which is what an instrumentation-heavy day looks like.

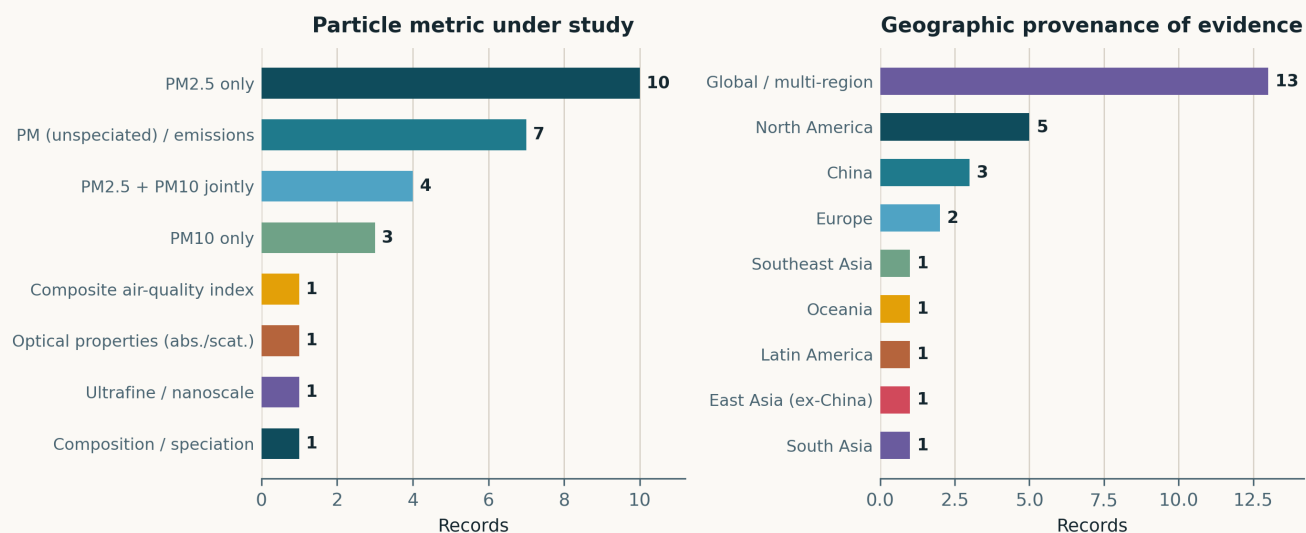


Fig. Left — particle metric. PM_{2.5} dominates, but note three records whose exposure metric is not a mass concentration at all (aerosol optical depth, absorbing aerosol index, ultrafine particle number) — these are the records that a mass-only monitoring programme cannot produce. Right — geographic provenance. No record today originates from sub-Saharan Africa, and the single South Asian record is a bench device study.

Life-course windows addressed today (records may span several stages)

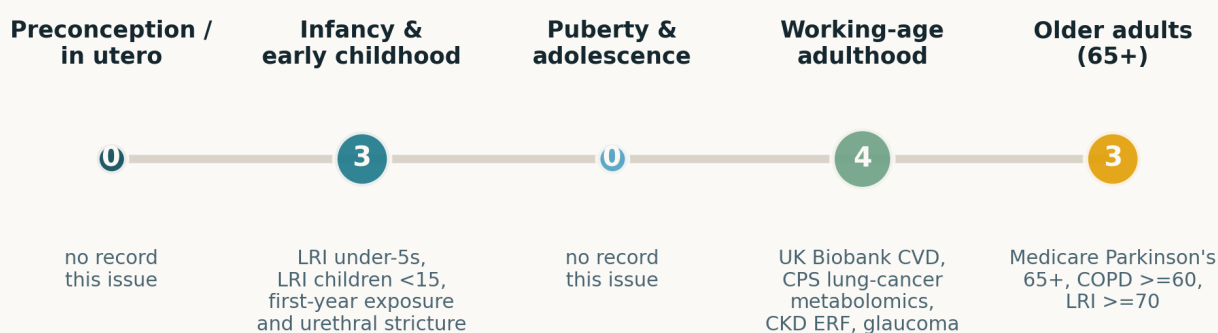


Fig. Life-course windows addressed. Counts are non-exclusive and describe this issue only. Both the in-utero and pubertal windows are empty today, an unusual gap given the last fortnight's developmental-origins run; the mass sits in working-age and 65+ cohorts.

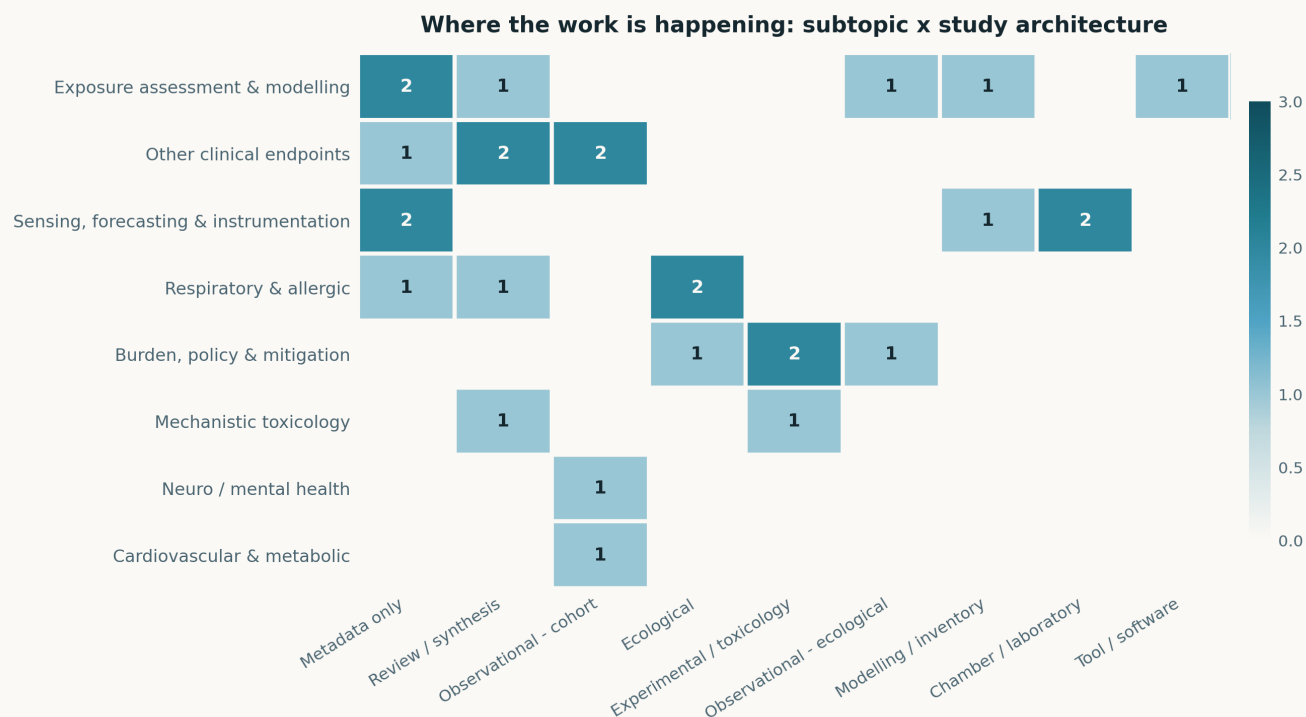


Fig. Subtopic $\times$ study-architecture cross-tabulation. The structural feature to note is the same one the 27 July issue flagged and which has not moved: the *Sensing* and *Exposure assessment* rows contain no health-endpoint linkage, and the health rows contain no instrumentation. The two literatures are running in parallel within this harvest.

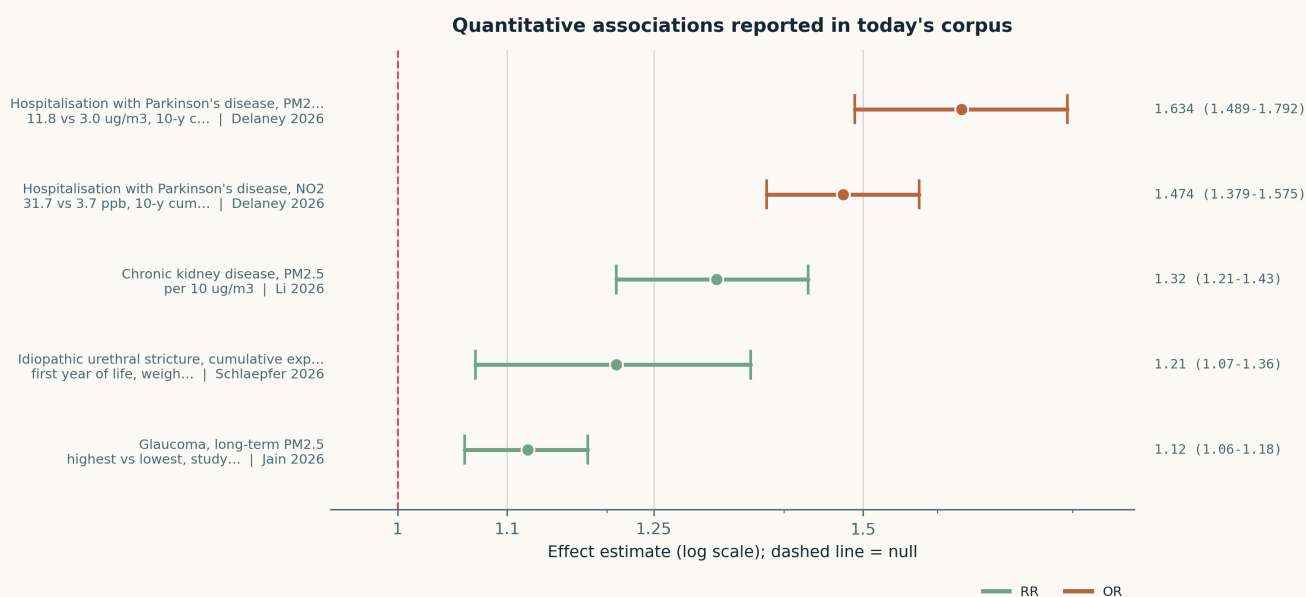


Fig. The five interval estimates carried today, on a common logarithmic axis. **Exposure contrasts are heterogeneous and the estimates are not mutually comparable:** Delaney's two ORs are 90th-vs-0.5th-percentile contrasts over a 10-year history, Li's RR is per $10 \mu\text{g}/\text{m}^3$, Jain's is a pooled highest-vs-lowest RR across studies with $I^2 = 78.7\%$, and Schlaepfer's is a weighted cumulative first-year index. The plot shows direction and precision only. Nothing here is pooled. Kakhia et al.'s counterfactual RR of 0.90 is omitted: the abstract reports a confidence interval on the risk difference and none on the ratio, and the projected-policy interval as printed (-1.05 ; CI -1.12 to -1.07) does not contain its own point estimate.

Paper-level digest

Ordered by cluster. Each entry gives the methodological core and the finding that matters, not an abstract paraphrase.

Neuro, cognition & mental health

1 record

Delaney et al. 2026 Environ Epidemiol

Cumulative annual air pollution exposure and first hospitalisation with Parkinson's disease

Discrete-time survival with distributed lag models over 10,366,083 Medicare FFS beneficiaries 65+, contiguous US 2000–2016; 10-year exposure histories for PM_{2.5}, NO₂ and summer O₃; non-linear exposure–response. **PM_{2.5} OR 1.634 (1.489–1.792)** at 11.8 vs. 3.0 µg/m³; **NO₂ OR 1.474 (1.379–1.575)** at 31.7 vs. 3.7 ppb; O₃ limited. Associations carried by exposure ≥4 years pre-event. The endpoint is hospitalisation, not incidence — diagnosis-to-hospitalisation lag is unmeasured and plausibly differential by exposure via healthcare access, which is the main threat to the latency claim. PM_{2.5} and NO₂ are strongly correlated in Medicare-scale data and the paper does not resolve which is doing the work.

doi: 10.1097/EE9.0000000000000519

Cardiovascular & metabolic

1 record

Kakhaia et al. 2026 Environ Int

Exposome approach to causal effects of time-varying air pollution mixtures on CVD incidence

UK Biobank, annual PM_{2.5} and NO₂ from a Europe-wide LUR model, 10-year cumulative CVD incidence under modified treatment policies. Confounders selected by a causal structure discovery algorithm with prior knowledge; sequential doubly robust non-parametric estimation with ensemble learning and censoring adjustment. A 25% exposure reduction over 2005–2019 gave –0.62 per 100 (95% CI –0.70 to –0.55), RR ≈ 0.90; the projected net-zero scenario reached –1.05 per 100 and RR ≈ 0.85. **The stated sensitivity is the finding:** dropping assessment-centre region from the adjustment set more than doubled the estimate, so the headline number is contingent on a spatial-confounding choice the authors cannot adjudicate. Method-forward, estimate-fragile.

doi: 10.1016/j.envint.2026.110441

Other clinical endpoints

5 records

Chow et al. 2026 Nat Commun

Blood metabolomic signatures linking air pollution to lung cancer

Prospective nested case-control across CPS-II and CPS-3, 1,357 participants, >1,100 metabolites in pre-diagnostic plasma, residential concentrations of six pollutants at blood draw. Eight metabolites associated with both exposure and subsequent lung cancer; four (γ-glutamylglutamine, phenylacetylglutamate, N-(2-furoyl)glycine, 4-vinylguaiacol glucuronide) significantly mediated PM and ozone associations at adjusted $q < 0.2$. Single-timepoint metabolomics cannot separate exposure response from early disease, and $q < 0.2$ is a permissive threshold over 1,100 features — treat the four as candidates. Smoking adjustment in a lung cancer metabolome study is the obvious residual-confounding exposure.

doi: 10.1038/s41467-026-75116-3

Li et al. 2026 Innovation (Camb)

Global and national CKD burden attributable to fine particles: updated exposure–response meta-analysis

Three-stage mixed-effects meta-analysis over 38 CKD and 16 eGFR studies, fitting a non-linear ERF and propagating it to 1990–2020 burden above a 10 µg/m³ counterfactual. **RR 1.32 (1.21–1.43) per 10 µg/m³**, predominantly log-linear; **eGFR –2.115 (0.452–3.777) mL/min/1.73 m²** with saturation at high exposure. 2020 burden: 485,000 deaths, 5.77 million incident and 243 million prevalent cases. The counterfactual floor is a choice, not a threshold, and the burden figures scale with it; the primary studies are also concentrated in high-exposure Asian settings, so the low-concentration arm of the ERF is thin.

doi: 10.1016/j.xinn.2026.101429

Jain et al. 2026 J Curr Glaucoma Pract

Ambient air pollution and glaucoma: systematic review and meta-analysis

PRISMA review, PubMed and Embase to 30 December 2025; 18 observational studies, 8 pooled by generic inverse variance. **Pooled RR 1.12 (1.06–1.18)** for long-term PM_{2.5}, $I^2 = 78.7\%$. NO₂, NO_x, PM₁₀ and CO directionally positive but synthesised narratively. RNFL and GCIPL thinning and modest IOP elevation also reported. Heterogeneity at 79% with eight studies makes the pooled point estimate a weak summary; the authors say causality cannot be established and they are right. Two databases only is a real recall limitation for this literature.

doi: 10.5005/jp-journals-10078-1522

Schlaepfer et al. 2026 Transl Androl Urol

Environmental hazards and the risk of idiopathic urethral stricture

Utah Population Database, 2,048 cases born 1996–2021 matched 1:1 on birth year and location to 1,939 controls; historical air pollution and toxics exposure assessed prenatally, in the first year of life, and in the five years before diagnosis. **Only the first-year window was associated: RR 1.21 (1.07–1.36)**. Univariate signals for ethylbenzene, tetrachloroethylene and molybdenum trioxide. Three exposure windows tested with one positive is a multiplicity problem the paper does not address, and the univariate toxicant hits are unadjusted. Included as a hypothesis, not a result.

doi: 10.21037/tau-2026-0338

Liu et al. 2026 JACC Adv

Reply: statistical fragility and methodological constraints in metabolomic mediation

Correspondence from the Emory group (Liu, Chang, Jones, Quyyumi, Liang) responding to a critique of metabolomic mediation between air pollution and cardiovascular risk. **Abstract not indexed** — carried as metadata only because it sits directly against Chow et al. (same senior author, same method class) in today's harvest and the methodological dispute is the reason to read both. Full text required.

doi: 10.1016/j.jacadv.2026.103149

Respiratory & allergic

4 records

Feng & Chao 2026 Int J Biometeorol*Lower respiratory infection burden attributable to PM_{2.5} in China and the G20, 1990–2023*

GBD 2023 deaths and DALYs from LRI attributable to ambient and household PM_{2.5}, Joinpoint regression on age-standardised DALY rates with sex subgroups and loess against SDI. Long decline reversed upward in 2021–2023 (APC **China 12.76–14.23%**, G20 **14.94–16.04%** for ambient); household-attributable rates kept falling in China but rose in the G20 from 2021 (APC 8.49–11.20%). Burden concentrated in under-5s and ≥70s. The post-COVID reversal is the interesting claim and it rests on three data-years of GBD modelled estimates, which are themselves smoothed — an inflection this recent should be treated as provisional.

doi: 10.1007/s00484-026-03283-5

Yang et al. 2026 J Thorac Dis*COPD and LRI burden attributable to household air pollution and secondhand smoke, 1990–2021*

GBD 2021 across 204 countries with BAPC and ARIMA projection. Age-standardised COPD DALYs and mortality fell 37.00% and 37.12%; LRI fell 66.35% and 53.62%. Burden stays concentrated in LMICs, in older women for COPD and in under-15s and ≥60s for LRI; HAP-attributable COPD declined faster than SHS-attributable. Projected female COPD burden from both sources exceeds male by 2030. Standard GBD secondary analysis — the exposure is a modelled solid-fuel-use proxy, not measured PM, and the projections inherit every assumption in the parent estimates.

doi: 10.21037/jtd-2026-0284

Wu et al. 2026 J Thorac Dis*Air pollution and COPD-related health outcomes: bibliometric and visualisation analysis*

11,167 Web of Science articles and reviews (1964 to 31 May 2025) through CiteSpace, VOSviewer and bibliometrix: country/institution collaboration, journal co-citation, keyword clustering, burst detection. USA, China and UK lead output; Harvard most influential institution; China and climate change identified as 2021-onward frontiers. Bibliometrics measure publication behaviour, not evidence — useful only for orienting a literature search or a grant narrative.

doi: 10.21037/jtd-2026-0784

The Lancet Respiratory Medicine 2026 Lancet Respir Med*The far-reaching effects of wildfire smoke*

Unsigned editorial. **Abstract not indexed**; carried as metadata only. Logged because wildfire-source attribution has been the recurring through-line of this watch since 27 July and an editorial in this journal signals where the clinical field is placing it. Full text required for any citable content.

doi: 10.1016/S2213-2600(26)00256-0

Mechanistic toxicology

2 records

Choi et al. 2026 Biofabrication*Degradable porous PLGA/PCL membrane enabling a lung alveoli-on-a-chip for particulate injury*

Porogen-assisted phase-separation membranes with tunable degradation (~40% surface porosity gain, ~3 μm thickness loss over culture) carrying primary human alveolar epithelium and microvascular endothelium at an air–liquid interface under breathing-mimetic cyclic strain. Apical diesel PM produced dose-dependent cytotoxicity, barrier permeability increase, elevated ROS and DNA damage *in both layers* — trans-barrier propagation, which an insert-only model cannot show. Roflumilast-N-oxide (PDE4 inhibitor) selectively attenuated oxidative and inflammatory readouts without restoring barrier integrity. Deposited dose is delivered apically in suspension and is not equivalent to inhaled dosimetry; no dose–response anchoring to ambient concentrations is attempted. Relevant here as a bench receptor for source-resolved particle samples.

doi: 10.1088/1758-5090/ae9402

Yuan et al. 2026 Innovation (Camb)*Hallmarks of lung cancer driven by inhalable particulate matter*

Critical review proposing a 12-feature “hallmarks of PM-associated lung cancer” framework spanning cell-intrinsic traits (genetic alteration, epigenetic reprogramming, metabolic remodelling, stemness) and microenvironment remodelling (inflammation, immune suppression, angiogenesis, pre-metastatic niche). Argues for integrating PM exposure into screening and risk stratification. A framework paper: the hallmark taxonomy is an organising device, not evidence, and the underlying studies are heterogeneous in particle source and dose. Useful as a citation scaffold for a mechanism section.

doi: 10.1016/j.xinn.2026.101347

Burden, policy & mitigation

4 records

Tian et al. 2026 Nanomaterials (Basel)*Bioinspired honeycomb-structured nanofibrous membranes for high-efficiency PM capture*

Template-assisted electrospinning onto a mesh receiver establishes a directional electric field, producing concentrated ordered stacking plus sparse random orientation of fine-diameter fibres. **>98.51% filtration efficiency at 34 Pa pressure drop**, 85% transparency, 130.8 mm/s air permeability. Efficiency is quoted for a single challenge condition; no loading curve, no electrostatic-decay ageing, no humidity sensitivity — the three things that determine whether an electret-assisted filter holds its number in service. Bench result, not a service life.

doi: 10.3390/nano16150941

Xie et al. 2026 J Thorac Dis*Thirty-year change and projection of Asian tracheal, bronchus and lung cancer burden*

GBD 2021 across Asian subregions with Joinpoint, age-period-cohort, decomposition, frontier, inequality and BAPC prediction. Incidence roughly doubled and deaths rose ~2.7-fold 1990–2021; ASIR +22%, ASPR +49%, ASMR +8%. Population growth was the largest contributor, then ageing. **Particulate matter pollution ranks second behind smoking among risk factors at ~10% and rising**. Six analytic frameworks on one modelled dataset multiply presentation, not information.

doi: 10.21037/jtd-2026-0740

Liu et al. 2026b Front Oncol*Lung cancer incidence and mortality trends with age-period-cohort analysis, Zibo 2014–2024*

Registry data for a heavily industrial Shandong city. Age-standardised incidence rose (AAPC **+4.34%**, 95% CI 2.14–6.59) while mortality fell (AAPC **–2.13%**, –3.53 to –0.72); male rates exceeded female throughout. Period effect increasing, cohort effect decreasing across successive birth cohorts. Rising incidence with falling mortality in a screening-active decade is the classic overdiagnosis signature and the paper does not test it; air pollution appears as narrative context with no exposure measurement at all.

doi: [10.3389/fonc.2026.1873592](https://doi.org/10.3389/fonc.2026.1873592)**Suryanarayanareddy et al. 2026** Environ Sci Pollut Res Int*Low-cost bio-based particle-capturing device for vehicular emissions: pilot study*

Cylindrical coconut-husk and sugarcane-pulp enclosure tapering to a conical outlet and a monitoring chamber, tested indoors under simulated junction conditions at three turbulence intensities. Single-pass removal 15–20% (PM_{2.5}) and 15–45% (PM₁₀) for halted vehicles; 20–65% and 15–65% for moving vehicles. Efficiency ordering flips with turbulence regime, which the paper reports honestly. Indoor simulation of an outdoor near-road environment, no replication count given, and no material-degradation testing on a biodegradable enclosure meant to sit at a roadside.

doi: [10.1007/s11356-026-38118-8](https://doi.org/10.1007/s11356-026-38118-8)

Exposure assessment & modelling

6 records

Park et al. 2026 Environ Pollut*City-scale non-exhaust PM₁₀ from emissions to exposure with a coupled chemistry–CFD framework*

Two non-exhaust emission-factor frameworks (EMEP/EEA Guidebook and KIMM) implemented in the Fine-scale Air Quality Simulation Unit, a 3-D coupled CFD–chemistry model, over a densely built Seoul district, combined with the 2022 Seoul Time-Activity Profile. **Total non-exhaust PM₁₀ was similar under both frameworks; source partitioning was not** — brake wear dominant under EMEP/EEA, road wear under KIMM. Non-exhaust reached **up to 70% of total PM₁₀** in near-road and dense cells; outdoor activity-weighted contributions of 9–10% in commercial and traffic zones exceeded concentration-based estimates because peak traffic coincides with peak outdoor presence. No measurement validation of the partitioning is presented, which is the obvious next step and the reason a receptor-model or tracer campaign would be complementary.

doi: 10.1016/j.envpol.2026.128941

Chillrud et al. 2026 Environ Epidemiol*pcpr: principal component pursuit for exposure pattern recognition and outlier detection*

R package implementing PCP with three environmental-health extensions: a non-negativity constraint on the low-rank matrix, missingness handling, and a specialised penalty for below-LOD observations. Demonstrated on 26 PM_{2.5} constituents measured every few days in Queens, NYC, 2015–2021: four patterns recovered (secondary/tailpipe, traffic, crustal dust, salt) plus **211 outlying events** including July Fourth firework spikes. PCP's sparse component is an outlier detector, not a source; calling the 211 events “exposure events” presumes an interpretation the decomposition does not supply. Still the most immediately usable artefact in today's corpus for anyone holding speciated network data.

doi: 10.1097/EE9.0000000000000515

Zhang et al. 2026 Environ Geochem Health*National-scale heavy metal contamination in urban road dust across 82 Chinese cities*

13,269 samples pooled from 169 studies into an urban-agglomeration framework; geo-accumulation index for anthropogenic enrichment, potential ecological risk index, multivariate source deconvolution. Cd and Hg are the priority pollutants, jointly **>85% of total ecological risk** in every agglomeration despite low absolute concentrations relative to Mn and Zn; a Cd/Hg I_{geo} ratio is proposed as a source fingerprint. Pooling 169 studies with heterogeneous sampling depth, particle-size cut and digestion protocol is the central weakness — the fingerprint may partly index methodology. Road dust is the resuspension reservoir behind non-exhaust PM, so this pairs with Park et al. above.

doi: 10.1007/s10653-026-03422-2

Florêncio et al. 2026 Atmos Environ*Comprehensive analysis of PM₁₀ in the Metropolitan Area of São Paulo, Part I: chemical characterisation and sources*

Abstract not deposited at Crossref on the entry date. Title indicates chemical speciation and source apportionment for a megacity with almost no component-resolved PM literature in this watch to date. Queued for the Saturday sweep.

doi: 10.1016/j.atmosenv.2026.122286

Stevens et al. 2026 Atmos Environ*Identifying unique biomass burning marker fingerprints for Australian hardwoods, softwoods and grasses*

Abstract not deposited at Crossref on the entry date. Fuel-specific tracer fingerprints are the missing input for wildfire source attribution outside the Northern Hemisphere; pairs with Pérez-Pastor et al. below on the metrology side. Queued for the Saturday sweep.

doi: 10.1016/j.atmosenv.2026.122303

Ibrahim 2026 Research Square (preprint)*Air pollution status and temporal patterns in Bukit Rambai and Muar, 2013–2017*

Five years of Malaysian Department of Environment station data (API, O₃, CO, SO₂, PM₁₀, NO₂) at two southern Peninsular stations, analysed in *openair* with calendar plots. Peak API 477 (Bukit Rambai) and **762 (Muar)** in June 2013 during a transboundary haze episode. Descriptive, two stations, no exposure or health linkage, and not peer reviewed. Carried only because episodic Southeast Asian haze is otherwise unrepresented in this watch and the 762 value is a useful anchor for dynamic-range requirements on low-cost optical sensors, which saturate well below it.

doi: 10.21203/rs.3.rs-10666765/v1

Sensing, forecasting & instrumentation

5 records

Kirchhoff et al. 2026 Atmos Meas Tech*Evaluation of DMSO as working fluid in condensation particle counters*

Laboratory and field comparison of a DMSO-operated CPC against a butanol counterpart across pressure, temperature and aerosol type, with a modified automatic refill system run for six months. Growth depends strongly on saturator temperature and saturator–condenser ΔT , confirmed by downstream optical particle counting and saturation/droplet-size simulation. **Mean cutoff 5.8 ± 0.9 nm (DMSO) vs. 5.6 ± 0.5 nm (butanol)**, with substantially lower fluid consumption; DMSO–H₂O mixtures extend the operating range. The wider cutoff spread on the DMSO unit is the honest cost, and matters if you are differencing two size channels. Non-flammability is what makes airborne and unattended remote deployment tractable.

doi: 10.5194/amt-19-5243-2026

Pérez-Pastor et al. 2026 Environ Sci Pollut Res Int*Robust quantification of anhydrosugars and reference material evaluation*

Accelerated solvent extraction (dichloromethane:methanol), BSTFA derivatisation, GC/MS for levoglucosan, galactosan and mannosan. Linearity $R > 0.9987$, repeatability $< 6\%$, LOD $< 0.02 \text{ mg L}^{-1}$; relative error 1–7.4% against SRM 2787 (fine PM). **SRM 1649b (urban dust) gave extraction-condition-dependent levoglucosan values**, which the authors attribute to matrix ageing — a direct warning against using an old urban-dust SRM to validate a biomass-burning tracer method. Chromatographic calibration dominates the uncertainty budget. Expanded uncertainties meet forthcoming EU air quality monitoring requirements.

doi: 10.1007/s11356-026-38136-6

Zhang et al. 2026b Atmos Meas Tech*Impact of aerosol-type assumption on Landsat 8 atmospheric correction over land*

600 Landsat 8 scenes over 100 AERONET sites, comparing the Urban Clean aerosol type hardwired into the LaSRC surface-reflectance algorithm against MOD04- and MOD09-based dynamic types. **MOD04 gave the best AOD accuracy: $R^2 = 0.7236$, RMSE 0.0437, bias 0.0052**. Relevant here not for surface reflectance but because the same fixed-aerosol-type assumption propagates into satellite AOD products used as predictors in satellite–surface PM_{2.5} fusion models — an R^2 of 0.72 against AERONET is the ceiling any such predictor inherits. AERONET sites are systematically non-representative of high-PM urban environments, so this is likely optimistic.

doi: 10.5194/amt-19-5281-2026

Cao et al. 2026 Atmos Environ*Characterising on-road NO₂ and particulate matter across urban road environments using city-scale mobile monitoring*

Abstract not deposited at Crossref on the entry date. The most directly relevant record of the day for mobile and distributed sensing and the one whose loss to the abstract famine costs most. Queued for the Saturday sweep.

doi: 10.1016/j.atmosenv.2026.122302

Zhang et al. 2026c Atmos Environ*Characterisation of air pollution transport using absorbing aerosol index from multiple payloads*

Abstract not deposited at Crossref on the entry date. Multi-payload AAI synergy is a transport-attribution method rather than a surface-concentration one; logged for completeness. Queued for the Saturday sweep.

doi: 10.1016/j.atmosenv.2026.122298

Corpus register

First author	Focus	Journal	Design	Metric	Region
Neuro / mental health					
Delaney et al.	Cumulative annual air pollution exposure and first hospitalization with Parkinson's disease	<i>Environ Epidemiol</i>	Registry cohort + geospatial	PM _{2.5} , NO ₂ , summer O ₃	United States
Cardiovascular & metabolic					
Kakhaia et al.	Causal effects of time-varying air pollution mixtures on cardiovascular disease incidence	<i>Environ Int</i>	Prospective cohort (mixtures)	PM _{2.5} , NO ₂	United Kingdom
Respiratory & allergic					
Feng & Chao	Lower respiratory infection burden attributable to PM _{2.5} in China and the G20, 1990-2023	<i>Int J Biometeorol</i>	GBD secondary analysis	Ambient PM _{2.5} , household PM _{2.5}	China
Lancet Respir Med	The far-reaching effects of wildfire smoke	<i>Lancet Respir Med</i>	Metadata only (no abstract)	Wildfire PM _{2.5}	Global
Wu et al.	Air pollution and COPD-related health outcomes: bibliometric and visualisation analysis	<i>J Thorac Dis</i>	Mapping review	PM _{2.5} , PM ₁₀ , household PM	Global
Yang et al.	COPD and LRI burden attributable to household air pollution and second-hand smoke, 1990-2021	<i>J Thorac Dis</i>	GBD secondary analysis	Household PM from solid fuel	Global
Mechanistic toxicology					
Choi et al.	Degradable PLGA/PCL membrane alveoli-on-a-chip for particulate-induced alveolar injury	<i>Biofabrication</i>	In vitro (cell lines)	Diesel particulate matter	In vitro
Yuan et al.	Hallmarks of lung cancer driven by inhalable particulate matter	<i>Innovation (Camb)</i>	Critical review	PM _{2.5} , PM ₁₀	Global
Sensing, forecasting & instrumentation					
Cao et al.	On-road NO ₂ and particulate matter across urban road environments from city-scale mobile monitoring	<i>Atmos Environ</i>	Metadata only (no abstract)	PM _{2.5} , NO ₂	Not stated
Kirchhoff et al.	Evaluation of DMSO as working fluid in condensation particle counters	<i>Atmos Meas Tech</i>	Laboratory metrology	Ultrafine particle number	Laboratory
Perez-Pastor et al.	Robust quantification of anhydrosugars and reference material evaluation for biomass burning tracers	<i>Environ Sci Pollut Res Int</i>	Laboratory metrology	PM biomass burning tracers	Spain
Zhang et al.	Impact of aerosol-type assumption on Landsat 8 atmospheric correction over land	<i>Atmos Meas Tech</i>	Model intercomparison	Aerosol optical depth	Global
Zhang et al.	Air pollution transport from absorbing aerosol index across multiple payloads	<i>Atmos Environ</i>	Metadata only (no abstract)	Absorbing aerosol index	Global
Exposure assessment & modelling					
Chillrud et al.	pcpr: principal component pursuit for exposure pattern recognition and outlier detection	<i>Environ Epidemiol</i>	Software / tool description	26 PM _{2.5} constituents	United States
Florencio et al.	Comprehensive analysis of PM ₁₀ in the Metropolitan Area of Sao Paulo, Part I	<i>Atmos Environ</i>	Metadata only (no abstract)	PM ₁₀	Brazil
Ibrahim	Air pollution status and temporal patterns in Bukit Rambai and Muar, 2013-2017	<i>Research Square (preprint)</i>	Ecological panel	PM ₁₀ , O ₃ , CO, SO ₂ , NO ₂	Malaysia
Park et al.	City-scale non-exhaust PM ₁₀ from emissions to exposure with a coupled chemistry-CFD framework	<i>Environ Pollut</i>	CTM simulation + source attribution	Non-exhaust PM ₁₀	Seoul
Stevens et al.	Biomass burning marker fingerprints for Australian hardwoods, softwoods and grasses	<i>Atmos Environ</i>	Metadata only (no abstract)	Biomass burning PM tracers	Australia
Zhang et al.	National-scale heavy metal contamination in urban road dust across 82 Chinese cities	<i>Environ Geochem Health</i>	Review / synthesis	Road dust heavy metals	China
Burden, policy & mitigation					
Liu et al.	Lung cancer incidence and mortality trends with age-period-cohort analysis, Zibo 2014-2024	<i>Front Oncol</i>	Ecological panel	Ambient air pollution (contextual)	China
Suryanarayanareddy et al.	Low-cost bio-based particle-capturing device for vehicular emissions: pilot study	<i>Environ Sci Pollut Res Int</i>	Bench / device experiment	PM _{2.5} , PM ₁₀ , CO, CO ₂	India
Tian et al.	Bioinspired honeycomb-structured nanofibrous membranes for high-efficiency PM capture	<i>Nanomaterials (Basel)</i>	Bench / device experiment	PM _{0.3} , PM _{2.5}	Laboratory
Xie et al.	Thirty-year change and projection of Asian tracheal, bronchus and lung cancer burden	<i>J Thorac Dis</i>	GBD secondary analysis	Ambient particulate matter pollution	Asia
Other clinical endpoints					

First author	Focus	Journal	Design	Metric	Region
Chow et al.	Blood metabolomic signatures linking air pollution to lung cancer	<i>Nat Commun</i>	Prospective cohort	PM _{2.5} , O ₃ , plus four co-pollutants	United States
Jain et al.	Ambient air pollution and glaucoma: systematic review and meta-analysis	<i>J Curr Glaucoma Pract</i>	Systematic review (PRISMA)	PM _{2.5} , PM ₁₀ , NO ₂ , NO _x , CO	Global
Li et al.	Global and national CKD burden attributable to fine particles: updated exposure-response meta-analysis	<i>Innovation (Camb)</i>	Umbrella review / meta-analysis	PM _{2.5}	Global
Liu et al.	Reply: statistical fragility and methodological constraints in metabolomic mediation	<i>JACC Adv</i>	Metadata only (no abstract)	PM _{2.5}	United States
Schlaepfer et al.	Environmental hazards and risk of idiopathic urethral stricture	<i>Transl Androl Urol</i>	Retrospective cohort	Ambient PM, VOCs, metal oxides	Salt Lake City

Provenance, screening and known gaps

Window. 2026-08-12 -> 2026-08-12, contiguous with the 11 August issue; no gap. PubMed is windowed on [Date - Entry]; the Crossref-by-ISSN leg is windowed on Crossref created. These are different clocks and are not treated as interchangeable.

Per-leg yield. PubMed connector, two queries (health axis, instrumentation axis): 19 distinct, of which 1 was returned by both — **18 carried**. Local harvester: pubmed_health 13 (5 additional records, 3 carried), pubmed_sensing 0, Europe PMC 2 (**2 carried**, one of them a Research Square preprint), Crossref by ISSN 16 (**7 carried**). arXiv returned 145 kB of relevance-ranked results with nothing on or after 5 August in scope. **OpenAlex HTTP 429, 10th consecutive** — the leg is dead and is not being retried beyond the single scripted call. Consensus returned 20 with the top 3 visible on the free tier; all three resolved to papers with no verifiable entry date in the window and were rejected rather than back-dated. **Sixth consecutive issue with zero Consensus carry.** Scholar Gateway's corpus ends May 2026 and cannot serve an August entry window; it was not used to fill the metadata-only records. ClinicalTrials.gov: no PM or air-pollution registration updated in the window; trials.json unchanged.

Rejections (14). Off-topic: dissolved black carbon photodegradation on hematite; meat sustainability labels; six ACP cloud/climate papers (ice supersaturation, CCN replenishment, Hunga ozone, aerosol-cloud in ECMWF, secondary ice production, sulphate AMOC response); N₂O clumped isotopes; IASI water vapour; stratospheric aerosol lidar. Out-of-axis: sulfur-VOC + OH kinetics (SOA precursor chemistry, no PM mass or health link); residential greenness and preeclampsia (exposure is NDVI, no PM estimate reported); TB recurrence in Asia (PM one determinant among many, and no DOI registered to verify). All logged with reasons to state/rejected.jsonl.

Metadata-only (6). Four Elsevier *Atmos. Environ.* articles, one Elsevier *JACC Adv* correspondence, one *Lancet Respir Med* editorial. These carry a title, journal and DOI and nothing else; every claim about them in this issue is about the title, and is marked as such. This is the fifth consecutive issue with a metadata-only block and the backlog now runs to 08-08.

Labels. Subtopic, design, particle-metric and geography labels are drawn only from the vocabularies already mapped in plots.py and mktable.py; no new leaf label was written this issue. Assignment is single-label by dominant contribution and is a judgement, not a controlled vocabulary.

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